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Boosting Organ Donation Self-Efficacy Through AI-Generated Health Messaging: The Differential Impacts of "Why" vs. "How" Messaging and Media Modality

Yoo Jung Oh [a], Sooyeon Park [b], and Ji Youn Ryu [c]

[a]Department of Communication, Michigan State University; [b]School of Communication, Florida State University; [c]School of Communication, The Ohio State University

ABSTRACT

Informed by construal level theory, this study used AI-generated health messages to examine how feasibility ("how") versus desirability ("why") framing, in either text or audiovisual modalities, influenced self-efficacy and organ donation sign-up intentions for individuals at different stages of behavior change. Results indicated that feasibility messages significantly boosted self-efficacy, especially among those who had not yet formed an intention, while media modality alone showed no significant effect. Self-efficacy was, in turn, positively associated with the likelihood to register as an organ donor. By integrating psychological theory with AI-generated health messaging, this study offers a cost-effective and scalable approach that can be adapted to various health domains. These insights are especially relevant for health communication professionals seeking to enhance audience engagement and promote critical health behaviors.

Posthumous organ donation is a life-saving act that is widely encouraged across many countries. However, the persistent shortage of registered organ donors remains a critical global health challenge. In the United States alone, 103,223 individuals are currently on the national transplant waiting list, and an average of 17 people die each day due to the lack of available organs (Health Resources and Services Administration [HRSA], 2025). The gap between the number of patients in need and available donors remains substantial, making it imperative to promote organ donor registration. Closing this gap requires a better understanding of the psychological processes underlying donor registration and the types of persuasive messages most likely to motivate action.

An important construct to target is self-efficacy, which refers to people's confidence in their ability to register as an organ donor. Lack of self-efficacy, especially regarding the perceived lack of knowledge on how to register as an organ donor, has been identified as a practical barrier that discourages people from signing up (Kaplow et al., 2024; Rocheleau, 2013). Unlike many other health behaviors (e.g., physical activity, healthy diet, quitting smoking), which require sustained motivation and ongoing self-regulation, registering as an organ donor is a one-time action. This means that a single successful enactment can ensure long-term behavioral completion (Hübner & Lippke, 2014), making self-efficacy a critical factor in the context of organ donation.

To develop campaign messages that effectively boost individuals' self-efficacy, this study draws on the construal level theory (CLT; Liberman & Trope, 2008; Trope & Liberman, 2010), which offers a valuable framework for guiding message design. For instance, the CLT argues that people formulate different levels of mental construal (high vs. low) when processing information. It suggests different messaging strategies to induce different levels of construal, such as emphasizing the desirability (why an action should be taken) that encourages high-level construal versus feasibility (how to take the action) that facilitates low-level construal. In promoting organ donation registration, feasibility messages may provide the concrete guidance needed to help potential donors register, whereas desirability messages may reinforce the moral and altruistic values underlying the decision.

To create these theory-based messages efficiently and consistently, this study employs generative artificial intelligence (AI) as a message production tool. Generative AI has been used in studies as a tool to create high-quality health messages in multiple formats, such as text and audiovisual content, while ensuring consistent adherence to theoretical guidelines. For example, Lim and Schmälzle (2023) employed prompt engineering to generate awareness messages about folic acid (a form of vitamin) and compared them to human-generated messages and found that AI-generated health messages were evaluated to be higher in quality and clarity. Beyond text generation, Lawson McLean and Gutiérrez Pineda (2024) demonstrated the use of AI-generated video for realistic medical training simulations, illustrating the technology's versatility in healthcare contexts. Despite the growing potential of AI in health promotion, research on AI-generated health messaging remains underdeveloped. Although AI can produce a wide range of health messages, few studies have applied established psychological frameworks to guide AI-assisted message creation and evaluate their persuasive effects.

CONTACT Yoo Jung Oh ohyoojun@msu.edu Department of Communication, Michigan State University, 404 Wilson Road, East Lansing, MI 48824
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Building on this potential, this study applies the CLT to test the effectiveness of theory-driven, AI-generated health messages in promoting organ donor registration. Here, we examine the effects of messaging strategy (emphasizing desirability vs. feasibility), stages of change (pre-intenders vs. intenders), and media modality (text vs. audiovisual) on organ donation self-efficacy, as well as how self-efficacy will affect intentions to sign up.

Literature review

Construal level theory

The CLT (Liberman & Trope, 2008; Trope & Liberman, 2010) explicates that individuals will process or think about an entity or an event differently depending on the perceived psychological distance with that entity/event. That is, individuals will formulate different levels of mental construal depending on how proximal or distant the entity/event is perceived to be. Specifically, people tend to think more *abstractly* about things that are psychologically *distant*, and more *concretely* about things that are psychologically *proximal*. This is based on the idea that human beings are egocentric, so anything that cannot be directly experienced is perceived distal (Liberman et al., 2007).

The CLT suggests that an entity/event is considered psychologically distant when it exists or takes place in the remote future (temporal), to people less like the self (social), in a distant place (spatial), and with a low likelihood (hypothetical). For example, if someone has a family member that signed up to be an organ donor, they are likely to feel closer to the topic of organ donation compared to someone who does not know any registered organ donor.

Based on this idea, the central proposition of the CLT is that the *greater* the perceived psychological distance, the more likely individuals are to use a *high-level* construal to process information in an abstract, idealized, and decontextualized manner; on the contrary, the *less* the perceived distance, the more likely they are to use a *low-level* construal to process the information in a concrete, specific, and contextualized manner. The relationship between psychological distance and mental construal has been shown in studies across different contexts (e.g., health risk, Ahn, 2014; climate change, Brügger et al., 2016; organ donation, Choi et al., 2011; decision making, Lu et al., 2013) as well as meta-analyses (Soderberg et al., 2015).

Feasibility versus desirability

The CLT suggests a variety of strategies in featuring high versus low construal in a persuasive message (see Lee, 2019; Soderberg et al., 2015 for detail). One popular strategy is to focus on either the *desirability* or *feasibility* of a target behavior. Desirability reflects the reasons for someone to engage in a behavior (i.e., "why"), whereas feasibility captures the specific ways to carry out the behavior (i.e., "how"). The CLT conceptualizes desirability as high-construal and feasibility as low-construal, since the former emphasizes abstract concepts like values, whereas the latter emphasizes more concrete details, which has been shown in previous studies (Freitas et al., 2004; Fujita et al., 2006). For instance, a desirability-focused organ donation campaign message would describe why someone should consider signing up, by elaborating the values of organ donation (e.g., "You can save lives") and the reward one can get by signing up (e.g., "Imagine the sense of fulfillment after signing up"). On the other hand, a feasibility-focused message would highlight how simple it is to sign up and provide details on how to sign up (e.g., "The sign-up process takes less than 5 minutes. Visit this website and fill out your information").

When it comes to encouraging organ donation sign-up, it is expected that a feasibility-focused message may be generally more effective in increasing self-efficacy. Although 90% of the U.S. population support organ donation (HRSA, 2025), only about 60% are registered donors. One common reason for this gap is lack of knowledge, such fears around organ transplantation procedure (Morgan et al., 2008) and mistrust of the medical system that doctors might not prioritize saving their lives if they are registered donors (Sanders, 2023). Given that, it is reasonable to think that a message that centers around delivering concrete information about the organ donation sign-up/transplant process (i.e., feasibility message) will be more effective in increasing self-efficacy compared to a message focusing on the values of organ donation (i.e., desirability message). Thus, we predict the following:

H1: Messaging strategy (desirability vs. feasibility) will impact self-efficacy such that participants in the feasibility message condition will report greater self-efficacy than those in the desirability message condition.

Matching psychological distance with construal level

The CLT suggests that the effectiveness of a persuasive message can be augmented when the construal level of the message is aligned with the psychological distance (Lutchyn & Yzer, 2011; Trope & Liberman, 2010). This is because matching the level of construal leads to consistent mental representations, which enables the message to be processed more fluently and increases message persuasiveness (Higgins, 2000; Lee, 2019). For example, for individuals who have high perceived temporal distance with the issue of climate change (e.g., "Climate change will happen in a distant future"), a desirability message (high construal) should be more effective than a feasibility message (low construal); on the contrary, for those who feel that the same climate issue is imminent (low temporal distance), a feasibility message should be more effective. This matching principle has been shown effective in increasing health outcomes in numerous previous studies (Bashir et al., 2014; Fujita et al., 2008; Mo et al., 2025; Saxton et al., 2024).

Applying this matching principle, we examine the effectiveness of desirability versus feasibility messages for individuals at either *pre- or post-intentional* phase of a behavior change. In so doing, we draw on the health action process approach (HAPA; Schwarzer, 1999, 2001), which is a dual-phase model that argues that the construction and implementation of health interventions must differ depending on whether individuals are at pre-intentional or post-intentional phase of behavior change. To specify, the HAPA proposes that for *pre-intenders* (i.e., those who have not yet developed their intention to

engage in a behavior), interventions should aim to increase their beliefs and perceptions to help them develop intentions. On the contrary, for *intenders* (i.e., those who have intentions to engage in a behavior), interventions must focus on facilitating the transformation of intention into actual behavior by informing (or encouraging individuals to think about) when, where, and how to perform a behavior (i.e., action planning), as well as ways to overcome potential barriers (i.e., coping planning). This is due to the fact that intenders, even though they have the motivation to carry out a behavior, often fail to do so due to the lack of skills and knowledge (Schwarzer et al., 2011).

Integrating the HAPA proposition into the CLT's matching principle, and building off of H2 (i.e., effectiveness of feasibility message), it can be argued that the effectiveness of feasibility message may be stronger for intenders compared to pre-inteders. In other words, although a feasibility message should be *generally* more effective than a desirability message (especially in the context of organ donation), intenders will benefit more from it than pre-intenders when it comes to increasing self-efficacy. This is because, first, intenders are likely to have less perceived psychological distance with organ donation sign-up compared to pre-intenders. Thus, based on the CLT's matching principle, the impact of a feasibility message (low construal) would be augmented for intenders (low distance). In alignment with this notion, Fujita et al. (2008) found that a feasibility message (i.e., focusing on steps to exercise) outperformed a desirability message (i.e., focusing on benefits of exercise) for temporally proximal health behaviors. Similarly, Bashir et al. (2014) reported that concrete messaging (i.e., how to reduce plastic use) increased pro-environmental intentions when participants perceived climate action to be happening in the near future. Relatedly, as previously mentioned, people who have intentions to become organ donors often fail to do so due to lack of knowledge about organ donation sign-up/transplant procedure (Morgan et al., 2008; Sanders, 2023). A feasibility message that provides a step-by-step guide to becoming an organ donor (e.g., how and where to sign up) will help intenders gain relevant knowledge and overcome potential barriers. Thus, we predict the following:

H2: The effect of the feasibility message on self-efficacy will be stronger for intenders than pre-intenders.

Effects of media modality: text versus audiovisual

The CLT has also been studied in the context of mode of delivery. Audiovisual representation provides clarity and concreteness regarding the objects of reference with diverse sensory cues, such as visual images, colors, background, sound, and tones (Amit et al., 2013; Choi et al., 2019). On the other hand, text compresses the attributes of an object, abstracting an object into its core and fundamental characteristics via symbolic representation. Through the lens of the CLT, text is perceived as more abstract (i.e., high construal) than audiovisual (i.e., low construal). Previous research suggests that multimedia health messages are more persuasive than text-based messages. Merion et al. (2003) specifically found that multimedia intervention is effective persuasive tools in organ donor registrations and family notification. Indeed, Yan et al. (2016) found that visual processing has more substantial capacity to stimulate behavioral intentions by facilitating a concrete mental imagery of the steps necessary to achieve a goal. In alignment with this, audiovisual health messages improve information retention compared to written messages (Bol et al., 2015). A meta analysis by Shen et al. (2015) further supports this, showing that audiovisual narratives have a larger effect on persuasion than textual narratives.

One possible explanation for these effects is that multimedia formats may support psychological mechanisms such as self-efficacy – individuals' belief in their ability to take specific action. By visualizing the actionable steps and consequences, video health messages might make health behaviors appear more approachable and manageable. Notably, Yeh et al. (2005) reported that self-efficacy is more strongly influenced by multimedia instructions than by written pamphlets. Based on these findings, we predict the following:

H3: Participants in the audiovisual message condition will report greater self-efficacy than those in the text message condition.

Matching media modality with mental construal

As discussed previously, the CLT argues that the message effectiveness can be maximized when the construal levels within a message are congruent. In support of this notion, Yan et al. (2016) found that individuals tended to rely more on visual processing (low construal) for psychologically proximal events, whereas they relied more on textual processing (high construal) for psychologically distant events. Moreover, they compared the effectiveness of desirability messages that differed in temporal distance (proximal vs. distal) and visualizability (visualizable vs. non-visualizable) and found that people preferred visualizable message (low construal) for temporally proximal event, whereas they preferred non-visualizable message (high construal) for distant event.

This suggests that a feasibility message may be better processed visually, whereas desirability messages may be better processed verbally. Indeed, Rim et al. (2015) showed that individuals relied on verbal processing when considering the reasons (i.e., desirability) for an action, but shifted to pictorial processing when carrying out the action (i.e., feasibility). Accordingly, we hypothesize the following interaction between media modality and messaging strategy:

H4: The effect of the feasibility message on self-efficacy will be stronger for participants in the audiovisual message condition than those in the text message condition.

Self-efficacy as a driver of sign-up intention

Finally, we examine how self-efficacy regarding organ donation will impact individuals' intentions to sign up as an organ donor. Self-efficacy is considered an influential factor in driving behavioral intention. Several prominent theories of behavior change (e.g., theory of planned behavior, Ajzen, 1991;

health belief model; Rosenstock, 2000; social cognitive theory; Bandura, 1986) highlight the role of self-efficacy as a key predictor of behavior or behavioral intention. The central idea of these theories is that one's confidence in their ability to manage or control situations influences their motivation to engage in a behavior (Bandura, 1986). Meta-analyses have shown a robust relationship between self-efficacy and behavior and/or behavioral intention (McEachan et al., 2011; Sheeran et al., 2016).

As discussed above, self-efficacy (i.e., one's confidence in their ability to sign up as a donor) is especially critical in the context of organ donation, given that lack of self-efficacy has been shown to be a barrier preventing people from becoming an organ donor (Kaplow et al., 2024; Rocheleau, 2013). In alignment with this notion, studies have shown that boosting self-efficacy effectively increased organ donation sign-up intentions (Hyde & White, 2009; Kong, 2022; Reubsaet et al., 2003; Steenaart et al., 2018). Thus, we predict the following:

H5: Self-efficacy will be positively related to intention to sign up for organ donation.

Figure 1 presents the proposed conceptual model along with the study hypotheses.

Method

Participants and design

Four hundred adults (50% Female, 49.8% Male, and 1% Non-binary; age $M = 28.02$, $SD = 15.63$, range from 18 to 80) from CloudResearch's Connect platform participated in this research. A 2 (Messaging strategy: Desirability vs. Feasibility) × 2 (Media modality: Text vs. Audiovisual) between-subject design was employed. Participants were randomly assigned one of the four conditions.

This study was reviewed by the university's Institutional Review Board (IRB) and determined to be exempt from additional IRB approval [No: 00010436]. Participants were presented with a consent page detailing the study's purpose, procedures, potential risks and benefits, and the voluntary nature of participation. Informed consent was obtained by all participants before proceeding with the study.

Procedure

Participants were asked to indicate their demographic information, stages of change (see Measures for detail), and AI perception prior to the stimuli. Before the exposure of stimuli, a disclosure regarding AI generated message was provided, "*Now, you will read and evaluate an organ donation campaign message. The following message was generated by an AI-based system, facilitated by ChatGPT-4 created by OpenAI.*" Subsequently, one of the four stimuli was displayed, followed by a post-questionnaire that measured the manipulation check items and study variables (self-efficacy and intentions).

Stimuli

Four versions of organ donation promotional messages that differed in messaging strategy (desirability vs. feasibility) and media modality (text vs. audiovisual) were generated by *ChatGPT-4* and *InVideo AI* in December 2024 (InVideo AI, 2024). The desirability message focused on the values of organ donation and reasons for signing up as an organ donor, whereas the feasibility message provided a step-by-step guide on how and where to sign up. Text stimuli were generated first, then audiovisual stimuli were generated based on the text stimuli (See Appendix A for prompts used to generate the messages and Appendix B for example stimuli). For the audiovisual stimuli, the authors carefully reviewed the stimuli to ensure consistency (e.g., same background music, caption font, and effects).

Measures

Stages of change

Based on Hall et al. (2007), participants were classified as either pre-intenders or intenders regarding organ donor registration

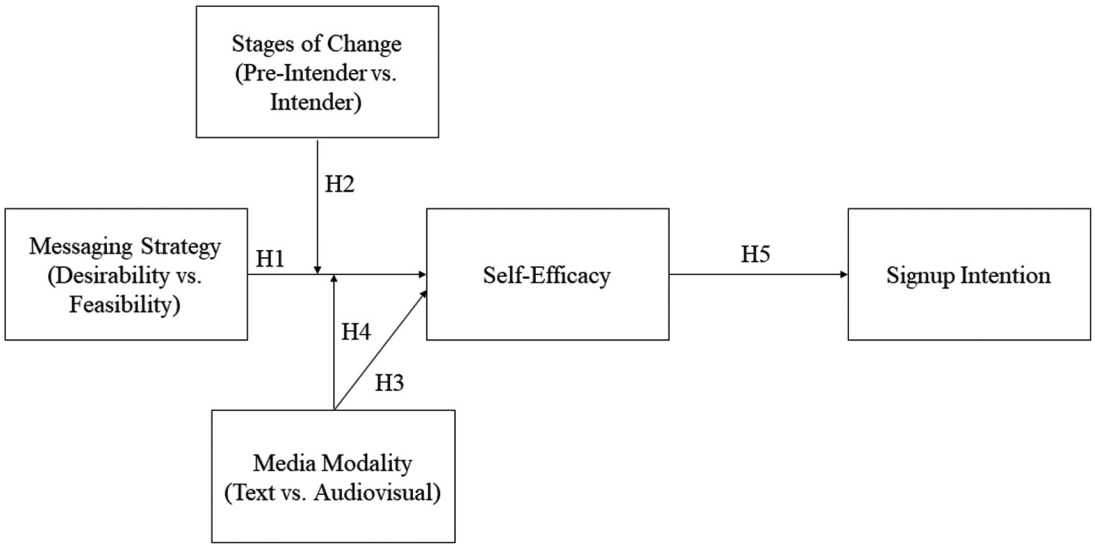


Figure 1. Proposed study model predicting self-efficacy and signup intention.

(see Figure 2 for a flowchart). Specifically, participants were initially categorized into five stages of change: precontemplation, contemplation, preparation, action, and maintenance. Then, drawing on the HAPA (Schwarzer, 2001; Schwarzer & Luszczynska, 2008), these stages were then grouped into two broader categories. The *intenders* group included participants in the maintenance ($n = 194$, 48.5%), action ($n = 19$, 4.8%), and preparation ($n = 19$, 4.8%) stages, totaling 232 individuals (58%). The *pre-intenders* group included those in the contemplation ($n = 39$, 9.8%) and precontemplation ($n = 129$, 32.3%) stages, totaling 168 individuals (42%).

Self-efficacy

Self-efficacy regarding organ donation sign up was adapted from previous research (Brug et al., 2000; Lemmens et al., 2005; Morgan & Miller, 2002; Robbins et al., 1999). This measure assessed individuals' perceived degree of confidence in their ability to sign up for organ donors registry (e.g., "If I wanted to, I could easily sign up on a donor registry.") on a 5-point Likert scale (1 = *I do not agree at all*, 5 = *I fully agree*), Cronbach's alpha = .82.

Intention to sign up

Intention to sign up for organ donation (Adapted from Masser et al., 2009) was measured with four items on a 5-point Likert scale (1 = *I do not agree at all*, 5 = *I fully agree*), e.g., "I would like to sign up for organ donation in the future," Cronbach's alpha = .99. See Table 1 for means, standard deviations, and correlations among all variables.

Results

Manipulation check

To examine whether participants in the desirability and feasibility conditions differed in their perceived construal level of the message, three items developed by the authors (e.g., "The content of the message emphasizes 'how' aspects of organ donation," 1 = *Strongly disagree*, 5 = *Strongly agree*) were used (Cronbach's alpha = .86). An independent samples t-test indicated that those in the feasibility message condition ($M = 4.11$, $SD = 0.70$) rated the message to be significantly greater in feasibility compared to those in the desirability message condition ($M = 3.16$, $SD = 1.02$), $t(398) = 10.93$, $p < .001$.

Hypotheses testing

A series of multiple hierarchical regression analyses were performed using SPSS 29.0 to test our study hypotheses (Figure 1). First, we examined the effects of messaging strategy, stages of change, and media modality on self-efficacy (H1–H4). Per Cohen et al. (2013), study variables (messaging strategy, stages of change, media modality) were entered in the first block, and their interaction terms were entered in the second block of the model. This stepwise approach was taken to assess the unique contribution of the main effects before evaluating whether the interaction terms accounted for additional variance in the outcome.

Regression analyses indicated that the model was significant, $F(3, 396) = 57.01$, $p < .001$, adj.$R^2 = .30$. Specifically,

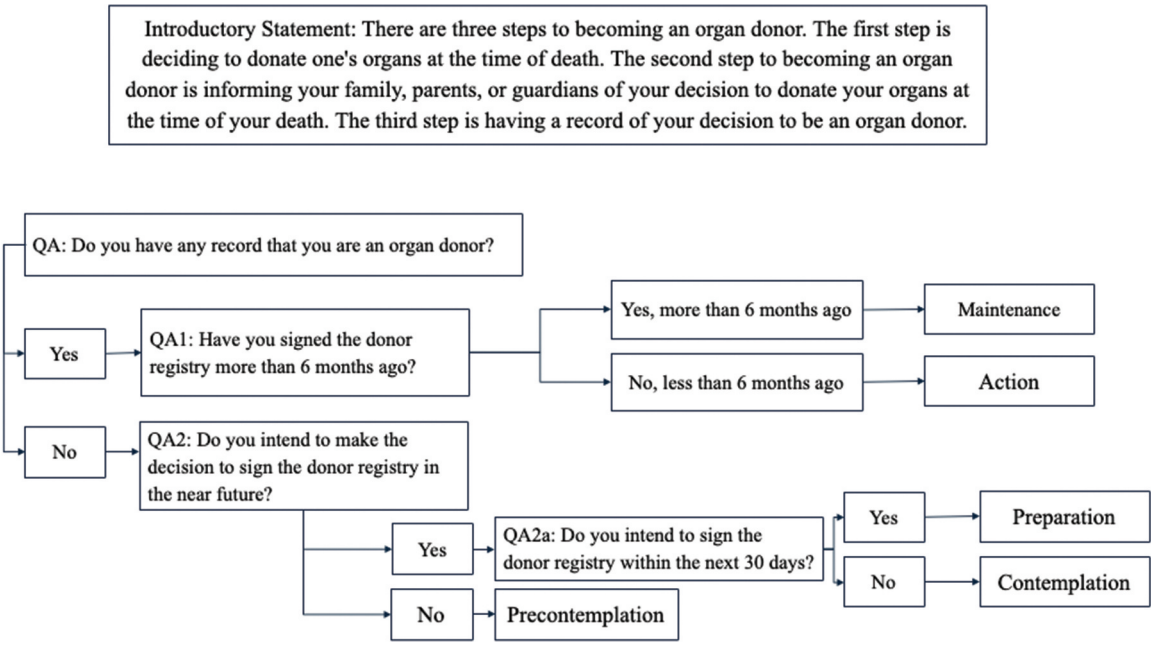


Figure 2. Stages of change algorithm for organ donor sign intention.

Table 1. Means, standard deviations, and correlations among study variables.

Variable	*M*	*SD*	1	2
1. Perceived Message Feasibility	3.65	0.99		
2. Self-efficacy	4.22	0.75	.22**	
3. Sign-up intention	3.52	1.28	.09*	.62**

Note. $N = 400$. *$p < .05$. **$p < .01$.

messaging strategy significantly and negatively predicted self-efficacy, $B = -0.21$, $SE = 0.06$, $p < .001$, such that those in the feasibility condition ($M = 4.21$, $SD = 0.67$) tended to report greater self-efficacy compared to those in the desirability condition ($M = 4.14$, $SD = 0.81$; H1 supported). However, media modality was not associated with self-efficacy ($p > .05$, H3 not supported), such that there was no significant difference in self-efficacy for text ($M = 4.22$, $SD = 0.73$) and audiovisual ($M = 4.22$, $SD = 0.77$) conditions. Although not hypothesized, stages of change significantly and positively predicted self-efficacy, such that intenders ($M = 4.56$, $SD = 0.52$) reported greater self-efficacy than pre-intenders ($M = 3.76$, $SD = 0.77$), $B = 0.81$, $SE = 0.06$, $p < .001$.

Next, the two-way interactions among study variables (block 2) significantly improved the model fit, $\Delta F(2, 394) = 4.21$, $p = .02$, $\Delta \text{adj.}R^2 = .02$. Specifically, there was a significant interaction between messaging strategy and stages of change, $B = 0.30$, $SE = 0.13$, $p = .02$. To probe this interaction, a simple effects analysis was conducted using the emmeans package (Lenth, 2024) in R studio 4.2. Inconsistent with H2, intenders' self-efficacy was not significantly different in the desirability ($M = 4.60$, $SE = 0.06$) and feasibility ($M = 4.52$, $SE = 0.06$) message conditions, $t(387) = 1.24$, $p = .22$, Cohen's $d = 0.12$. On the contrary, for pre-intenders, those in the feasibility message condition ($M = 3.94$, $SE = 0.07$) reported significantly greater self-efficacy than those who read desirability message ($M = 3.55$, $SE = 0.07$), $t(387) = 3.46$, $p < .001$, $d = 0.34$. That is, data were in the opposite direction of our prediction, such that the effect of the feasibility message was stronger for *pre-intenders* (See Figure 3 for simple slope visualization). Additionally, no significant messaging strategy-media modality interaction on self-efficacy was detected ($p > .05$, H4 not supported).

Lastly, we assessed the relationship between self-efficacy and organ donation sign-up intention (H5). The model was statistically significant, $F(1, 398) = 251.93$, $p < .001$, $\text{adj.}R^2 = .39$. In support of H5, self-efficacy was positively associated with sign-up intention, $B = 1.06$, $SE = 0.07$, $p < .001$ (see Table 2).

Post-hoc, we conducted an additional moderated mediation analysis to examine the indirect effects of messaging strategy (moderated by media modality and stages of change) on sign-up intentions via self-efficacy. See Appendix C for details.



Figure 3. Simple slope analysis visualization.

Table 2. Hierarchical regression results.

	Outcome Variable: Self-efficacy				
	B	*SE*	β	*t*	*sr*
First Block (Study Variables)					
Messaging Strategy	−0.21	0.06	−.14	−3.36***	−.14
Stages of Change	0.81	0.06	.54	12.77***	.54
Modality	0.01	0.06	.01	0.15	.01
	F_{change} (3, 396) = 57.01, $p < .001$, $\text{adj.}R^2_{change} = .30$				
Second Block (Interaction)					
Messaging Strategy*Stages of Change	0.30	0.13	.18	2.35*	.10
Messaging Strategy*Modality	−0.21	0.13	−.12	−1.68	−.07
	F_{change} (2, 394) = 4.21, $p = .02$, $\text{adj.}R^2_{change} = .02$				
	Outcome Variable: Sign-up Intention				
Self-efficacy	1.06	0.07	.62	15.87***	.62
	$F(1, 398) = 251.93$, $p < .001$, $\text{adj.}R^2 = .39$				

*$p < .05$. ***$p < .001$.

Discussion

Based on the CLT's matching principle, this study used an experimental approach to examine the impact of messaging strategy (desirability vs. feasibility) and media modality (text vs. video) on individuals' self-efficacy regarding organ donation and intentions to sign up to be an organ donor. Drawing on the HAPA model, we further examined whether the impact of messaging strategy would differ depending on the stages of behavior change (pre- vs. post-intentional) individuals are at.

As predicted, results showed that those who read a feasibility message focusing on "how" to sign up reported greater self-efficacy compared to those who read a desirability message focusing on "why" one should sign up. Additionally, there was an interaction between messaging strategy and stages of change, such that the feasibility message was more effective for *pre-intenders*, which was the opposite of our prediction that it would be more effective for intenders. Furthermore, no significant effect of message modality on self-efficacy was detected, nor its interaction with stages of change. Finally, in alignment with previous literature, self-efficacy was positively related to intentions to sign up.

Implications

This study offers both theoretical and practical contributions by demonstrating how the application of the CLT propositions into AI-generated health messages can enhance their effectiveness. AI's versatility allows for rapid creation of tailored messages through simple prompts that align with theory (e.g., the CLT's high- vs. low-construal framing), thus augmenting the effectiveness of persuasive messages for individuals at various stages of behavior change.

Our finding that the feasibility message (explaining "how" to sign up as an organ donor) was more effective than the desirability message (focusing on "why" one should become an organ donor) in increasing self-efficacy even among the pre-intenders, adds nuance to the CLT's matching principle by suggesting it may operate differently in high-consensus attitude contexts (i.e., when people generally hold positive attitudes toward organ donation). This indicates the importance of considering baseline societal attitudes when predicting whether high- or low-construal messages will be more effective. Integrating such contextual factors into the CLT framework could help refine its predictive power in health communication research.

As for practical implications, our findings highlight the importance of providing concrete and detailed information on organ donor registration processes, rather than simply emphasizing the values of becoming an organ donor. The vast majority of the U.S. citizens already hold positive attitudes toward organ donation (HRSA, 2025); yet many remain unregistered due to a lack of knowledge (Morgan et al., 2008) and mistrust of the medical system (Sanders, 2023). In such cases, building confidence may depend more on reducing logistical uncertainty than on reinforcing moral or altruistic arguments. Practical ways to achieve this goal include making accessible step-by-step guides of the sign-up process to facilitate the transformation of positive attitudes into behavior and providing concrete facts about the transplantation process to help people overcome vague fear about organ donation. This aligns with Morgan et al.'s (2003) argument that organ donation promotion messages should center around building knowledge of organ donation and countering rumors surrounding organ donation.

Our results revealed that intenders generally show higher self-efficacy than pre-intenders, but the effect of messaging strategy (desirability vs. feasibility) on self-efficacy was not significant. This could mean that there was a ceiling effect for intenders (already having relatively high self-efficacy), and thus neither type of message had limited additional impact on enhancing their self-efficacy. On the other hand, there was a significant effect of feasibility messages on self-efficacy for pre-intenders. While unexpected, this finding is not entirely surprising. Given that the majority of U.S. citizens currently hold positive attitudes toward organ donation (HRSA, 2025), it is reasonable to assume that even pre-intenders are likely to hold favorable perceptions of organ donation and already acknowledge the value (i.e., desirability) of signing up as an organ donor. Therefore, feasibility messages that provide tangible steps to register as organ donor may have contributed to increasing pre-intenders' self-efficacy, rather than desirability messages highlighting the value and importance of organ donation which they may have been already aware of.

Inconsistent with our prediction that the audiovisual message would be more effective than the text message in encouraging self-efficacy, no significant effect of media modality was found. This result is not totally surprising given previous findings (e.g., Graaf et al., 2016; Winterbottom et al., 2012). For example, Graaf et al. (2016) reported that the medium itself did not significantly influence behavioral intention, suggesting that media modality may not consistently impact persuasiveness in health-related contexts. Similarly, Winterbottom et al. (2012) found that narratives delivered in both written and video formats significantly influenced participants' decisions regarding dialysis modality, though no differences emerged between the two formats. Prior research, however, suggests that multisensory elements can enhance message effectiveness (Xu, 2017).

Finally, these results underscore the growing potential of AI to serve as a scalable tool for theory-driven health communication. By embedding theoretical constructs into AI prompts, practitioners can generate tailored messages with minimal time and cost investment. This approach could accelerate message testing and allow for rapid iteration based on audience feedback, while maintaining theoretical rigor. While this study examined only AI-generated messages, future work could compare AI-generated messages with human-generated messages to test whether the same theoretical framing would produce comparable effects. Such research would clarify whether observed effects are driven primarily by message content or moderated by perceptions of the source.

Limitations

Although this study provides valuable insights into persuasive health message design, it is not without limitations. First, due to the cross-sectional nature of this study, we

did not measure actual behavior following behavioral intention. Although intention is considered one of the most proximal predictors of actual behavior (Ajzen, 1991), some scholars have pointed out that intention may not always be translated into behavior (Sheeran et al., 2016). Future studies must address this limitation by measuring actual sign-up behavior.

Second, despite the increased use of AI in persuasive message generation, the *Black-box problem* of AI persists (Von Eschenbach, 2021). The Black box problem of AI, which refers to the difficulty in understanding the decision-making processes and the internal workings of AI, has been pointed out as one reason for decreased trust in AI-generated messages (Von Eschenbach, 2021; Wadden, 2022). Relatedly, it has been shown that AI systems can sometimes generate inaccurate or misleading content (i.e., hallucination; Beutel et al., 2023), which may pose risks to individuals who may rely on this information for making important health decisions. Thus, to prevent such unintended harms and increase transparency underlying the message generation process, the role of health communicators is critical; specifically, they should engage in robust monitoring and come up with review protocols to validate the accuracy of the AI-generated messages prior to dissemination. Additionally, maintaining transparency by sharing prompts can help safeguard against inaccuracies in AI-generated messages and increase reproducibility of the content.

Conclusion

Leveraging the CLT to design AI-driven organ donation-related health messages, our study shows strong preliminary evidence for increasing self-efficacy and intention to sign up on the organ donor registry. AI-based message creation thus holds promise for improving the effectiveness and reach of organ donation efforts, paving the way for continued innovation and research in this area.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Yoo Jung Oh http://orcid.org/0000-0002-7829-8535
Sooyeon Park http://orcid.org/0000-0002-4580-6775
Ji Youn Ryu http://orcid.org/0000-0002-5408-0507

Data availability statement

Available upon request.

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